

Zeta-Law Entropy, Modular Resolution, and Three Applications to Zeta Inequalities

1 The zeta law as a Gibbs law

All logarithms are natural. For $\beta > 1$, define the Riemann zeta probability law

$$\rho_\beta(n) := \frac{n^{-\beta}}{\zeta(\beta)}, \quad n \in \mathbb{N}.$$

Equivalently,

$$\rho_\beta(n) = \frac{e^{-\beta E(n)}}{Z(\beta)}, \quad E(n) = \log n, \quad Z(\beta) = \zeta(\beta).$$

Thus the Riemann zeta function is the partition function of the energy landscape $E(n) = \log n$ on $\mathbb{N}$.

We use the notation

$$A(\beta) := \log \zeta(\beta).$$

Then A is the free energy of the zeta law. Its first two derivatives are

$$A'(\beta) = -\mathbb{E}_\beta[\log N],$$

and

$$A''(\beta) = \text{Var}_\beta(\log N).$$

More generally, the derivatives of A are cumulants of $\log N$ under ρ_β .

Proposition 1 (Zeta-law calculus). *Let $\beta > 1$ and let N be distributed according to ρ_β . Then:*

$$\mathbb{E}_\beta[N^{-r}] = \frac{\zeta(\beta + r)}{\zeta(\beta)} \quad (r \geq 0).$$

Moreover, for $\alpha, \beta > 1$,

$$D(\rho_\alpha \| \rho_\beta) = (\beta - \alpha) \mathbb{E}_\alpha[\log N] + \log \zeta(\beta) - \log \zeta(\alpha).$$

Finally,

$$(\log \zeta)''(\beta) = \text{Var}_\beta(\log N) \geq 0.$$

Thus $\log \zeta$ is convex on $(1, \infty)$.

Proof. The moment identity is immediate:

$$\mathbb{E}_\beta[N^{-r}] = \sum_{n=1}^{\infty} \frac{n^{-\beta}}{\zeta(\beta)} n^{-r} = \frac{\zeta(\beta + r)}{\zeta(\beta)}.$$

For the relative entropy,

$$\begin{aligned}
D(\rho_\alpha \parallel \rho_\beta) &= \sum_{n=1}^{\infty} \rho_\alpha(n) \log \frac{\rho_\alpha(n)}{\rho_\beta(n)} \\
&= \sum_{n=1}^{\infty} \rho_\alpha(n) [(\beta - \alpha) \log n + \log \zeta(\beta) - \log \zeta(\alpha)] \\
&= (\beta - \alpha) \mathbb{E}_\alpha[\log N] + \log \zeta(\beta) - \log \zeta(\alpha).
\end{aligned}$$

Differentiating $A(\beta) = \log \zeta(\beta)$ gives

$$A'(\beta) = \frac{\zeta'(\beta)}{\zeta(\beta)} = -\mathbb{E}_\beta[\log N],$$

and

$$A''(\beta) = \frac{\zeta''(\beta)}{\zeta(\beta)} - \left(\frac{\zeta'(\beta)}{\zeta(\beta)} \right)^2 = \mathbb{E}_\beta[(\log N)^2] - \mathbb{E}_\beta[\log N]^2.$$

□

2 Successor entropy and modular resolution

For $q \geq 2$, define the residue distribution of the zeta law modulo q by

$$\mu_{q,\beta}(a) := \sum_{\substack{n \geq 1 \\ n \equiv a \pmod{q}}} \rho_\beta(n), \quad a \in \mathbb{Z}/q\mathbb{Z}.$$

We write $a + 1$ cyclically in $\mathbb{Z}/q\mathbb{Z}$ and define the modular successor entropy

$$B_q(\beta) := \sum_{a \in \mathbb{Z}/q\mathbb{Z}} \mu_{q,\beta}(a) \log \frac{\mu_{q,\beta}(a)}{\mu_{q,\beta}(a+1)}.$$

Theorem 2 (Zeta-law successor entropy). *For every $\beta > 1$,*

$$\Sigma(\beta) := \sum_{n=1}^{\infty} \rho_\beta(n) \log \frac{\rho_\beta(n)}{\rho_\beta(n+1)} = \frac{\beta}{\zeta(\beta)} \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \zeta(\beta + k). \quad (1)$$

Furthermore,

$$\Sigma(\beta) = \sup_{q \geq 2} B_q(\beta) = \lim_{q \rightarrow \infty} B_q(\beta). \quad (2)$$

Equivalently,

$$\frac{\beta}{\zeta(\beta)} \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \zeta(\beta + k) = \sup_{q \geq 2} \sum_{a \in \mathbb{Z}/q\mathbb{Z}} \mu_{q,\beta}(a) \log \frac{\mu_{q,\beta}(a)}{\mu_{q,\beta}(a+1)}. \quad (3)$$

Proof. Since

$$\frac{\rho_\beta(n)}{\rho_\beta(n+1)} = \left(1 + \frac{1}{n}\right)^\beta,$$

we have

$$\Sigma(\beta) = \frac{\beta}{\zeta(\beta)} \sum_{n=1}^{\infty} n^{-\beta} \log \left(1 + \frac{1}{n}\right).$$

Using

$$\log(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} x^k \quad (0 < x \leq 1)$$

and monotone Abel limiting at $x = 1$ gives

$$\Sigma(\beta) = \frac{\beta}{\zeta(\beta)} \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \zeta(\beta+k).$$

It remains to prove the modular variational identity. Fix $q \geq 2$. For each residue class a , the log-sum inequality gives

$$\sum_{\substack{n \geq 1 \\ n \equiv a \pmod{q}}} \rho_{\beta}(n) \log \frac{\rho_{\beta}(n)}{\rho_{\beta}(n+1)} \geq \mu_{q,\beta}(a) \log \frac{\mu_{q,\beta}(a)}{\sum_{n \equiv a \pmod{q}} \rho_{\beta}(n+1)}.$$

For $a \neq 0$, the denominator is $\mu_{q,\beta}(a+1)$. For $a = 0$, the denominator is $\mu_{q,\beta}(1) - \rho_{\beta}(1)$, which is at most $\mu_{q,\beta}(1)$. Therefore

$$\Sigma(\beta) \geq B_q(\beta) \quad (q \geq 2),$$

and so

$$\Sigma(\beta) \geq \sup_{q \geq 2} B_q(\beta).$$

Conversely, fix $M \in \mathbb{N}$. For $q > M+1$, the residue classes $1, 2, \dots, M+1$ isolate the first $M+1$ integers up to tails of size $O_{\beta}(q^{-\beta})$:

$$\mu_{q,\beta}(a) = \rho_{\beta}(a) + O_{\beta,M}(q^{-\beta}) \quad (1 \leq a \leq M+1).$$

Also,

$$\mu_{q,\beta}(0) = q^{-\beta}.$$

For $1 \leq a \leq q-2$, the sequence $\mu_{q,\beta}(a)$ is decreasing in a , and

$$\mu_{q,\beta}(q-1) > \mu_{q,\beta}(0).$$

Hence all cyclic terms in $B_q(\beta)$ are nonnegative except possibly the boundary term

$$\mu_{q,\beta}(0) \log \frac{\mu_{q,\beta}(0)}{\mu_{q,\beta}(1)} = O_{\beta}(q^{-\beta} \log q),$$

which tends to 0. Therefore

$$\liminf_{q \rightarrow \infty} B_q(\beta) \geq \sum_{n=1}^M \rho_{\beta}(n) \log \frac{\rho_{\beta}(n)}{\rho_{\beta}(n+1)}.$$

Letting $M \rightarrow \infty$ yields

$$\liminf_{q \rightarrow \infty} B_q(\beta) \geq \Sigma(\beta).$$

Together with $B_q(\beta) \leq \Sigma(\beta)$, this proves

$$\lim_{q \rightarrow \infty} B_q(\beta) = \Sigma(\beta),$$

and hence the supremum identity. □

Corollary 3 (Prime-modulus Dirichlet L -resolution). *Let p be prime and let χ range over Dirichlet characters modulo p . For $a \in \{1, \dots, p-1\}$, define*

$$S_a(p, \beta) := \sum_{\chi \bmod p} \overline{\chi(a)} L(\beta, \chi).$$

Then

$$\mu_{p,\beta}(a) = \frac{S_a(p, \beta)}{(p-1)\zeta(\beta)} \quad (1 \leq a \leq p-1),$$

whereas

$$\mu_{p,\beta}(0) = p^{-\beta}.$$

Consequently,

$$\Sigma(\beta) \geq F_p(\zeta(\beta), \{L(\beta, \chi)\}_{\chi \bmod p}) \quad (p \text{ prime}, \beta > 1). \quad (4)$$

where

$$\begin{aligned} F_p &= p^{-\beta} \log \frac{p^{-\beta}(p-1)\zeta(\beta)}{S_1(p, \beta)} \\ &+ \frac{1}{(p-1)\zeta(\beta)} \sum_{a=1}^{p-2} S_a(p, \beta) \log \frac{S_a(p, \beta)}{S_{a+1}(p, \beta)} \\ &+ \frac{S_{p-1}(p, \beta)}{(p-1)\zeta(\beta)} \log \frac{S_{p-1}(p, \beta)}{p^{-\beta}(p-1)\zeta(\beta)}. \end{aligned}$$

Moreover,

$$\Sigma(\beta) = \lim_{\substack{p \rightarrow \infty \\ p \text{ prime}}} F_p(\zeta(\beta), \{L(\beta, \chi)\}_{\chi \bmod p}) = \sup_{p \text{ prime}} F_p(\zeta(\beta), \{L(\beta, \chi)\}_{\chi \bmod p}). \quad (5)$$

Proof. For $a \not\equiv 0 \pmod{p}$, character orthogonality gives

$$\frac{1}{p-1} \sum_{\chi \bmod p} \overline{\chi(a)} \chi(n) = \begin{cases} 1, & n \equiv a \pmod{p}, \\ 0, & n \not\equiv a \pmod{p}, \end{cases}$$

for n coprime to p . Hence

$$\sum_{\substack{n \geq 1 \\ n \equiv a \pmod{p}}} n^{-\beta} = \frac{1}{p-1} \sum_{\chi \bmod p} \overline{\chi(a)} L(\beta, \chi).$$

Dividing by $\zeta(\beta)$ gives the formula for $\mu_{p,\beta}(a)$. The zero class satisfies

$$\mu_{p,\beta}(0) = \frac{1}{\zeta(\beta)} \sum_{m=1}^{\infty} (mp)^{-\beta} = p^{-\beta}.$$

Substituting these expressions into $B_p(\beta)$ gives F_p . The inequality and limiting identity follow from Theorem 2 along the prime sequence $p \rightarrow \infty$. $\square$

Remark 4 (Interpretation). *Theorem 2 says that the microscopic additive successor entropy of the zeta law,*

$$n \mapsto n+1,$$

is recovered exactly from its finite modular shadows,

$$n \mapsto n \bmod q.$$

For prime moduli, the modular shadows are finite nonlinear functionals of Dirichlet L -values. Thus an additive entropy production observable of ζ is resolved through Dirichlet spectral data.

3 Euler-score observations

The modular entropy theorem uses the residue shadows

$$\mu_{q,\beta}(a) = \mathbb{P}_\beta(N \equiv a \pmod{q})$$

of the zeta law. A second, Euler-product shadow is obtained by looking only at the zero residue class:

$$\mu_{d,\beta}(0) = \mathbb{P}_\beta(d \mid N) = d^{-\beta}.$$

These zero-class shadows decompose the zeta energy.

Proposition 5 (Euler-score decomposition). *For every $\beta > 1$,*

$$-(\log \zeta)'(\beta) = \mathbb{E}_\beta[\log N] = \sum_{d=1}^{\infty} \Lambda(d) \mu_{d,\beta}(0) = \sum_{d=1}^{\infty} \frac{\Lambda(d)}{d^\beta}.$$

Equivalently,

$$-\frac{\zeta'(\beta)}{\zeta(\beta)} = \sum_{d=1}^{\infty} \frac{\Lambda(d)}{d^\beta}.$$

Proof. The identity

$$\log n = \sum_{d \mid n} \Lambda(d)$$

gives

$$\begin{aligned} \mathbb{E}_\beta[\log N] &= \sum_{n=1}^{\infty} \rho_\beta(n) \sum_{d \mid n} \Lambda(d) \\ &= \sum_{d=1}^{\infty} \Lambda(d) \mathbb{P}_\beta(d \mid N). \end{aligned}$$

But

$$\mathbb{P}_\beta(d \mid N) = \frac{1}{\zeta(\beta)} \sum_{m=1}^{\infty} (dm)^{-\beta} = d^{-\beta}.$$

Thus

$$\mathbb{E}_\beta[\log N] = \sum_{d=1}^{\infty} \frac{\Lambda(d)}{d^\beta}.$$

Since

$$\mathbb{E}_\beta[\log N] = -(\log \zeta)'(\beta),$$

the claim follows. □

Proposition 6 (Finite Euler-score identity). *For every $A \subseteq \{1, \dots, N\}$,*

$$\sum_{n \in A} \log n = \sum_{d \leq N} \Lambda(d) \sum_{m \leq N/d} 1_A(md).$$

Proof. Reverse the order of summation:

$$\begin{aligned} \sum_{d \leq N} \Lambda(d) \sum_{m \leq N/d} 1_A(md) &= \sum_{n \in A} \sum_{d \mid n} \Lambda(d) \\ &= \sum_{n \in A} \log n. \end{aligned}$$

□

4 A curvature lemma for the cold zeta law

The following estimate is the free-energy curvature input used in the second application.

Lemma 7 (Uniform positive-axis curvature bound). *For every $s \geq 3$,*

$$(\log \zeta)''(s) = \text{Var}_s(\log N) < \frac{1}{3}. \quad (6)$$

Proof. By Proposition 1,

$$(\log \zeta)''(s) = \frac{\zeta''(s)}{\zeta(s)} - \left(\frac{\zeta'(s)}{\zeta(s)} \right)^2 \leq \frac{\zeta''(s)}{\zeta(s)} \leq \zeta''(s).$$

For $s \geq 3$,

$$\zeta''(s) = \sum_{m=2}^{\infty} \frac{(\log m)^2}{m^s} \leq \sum_{m=2}^{\infty} \frac{(\log m)^2}{m^3}.$$

The function $x \mapsto (\log x)^2 x^{-3}$ is decreasing for $x \geq 2$. Therefore

$$\sum_{m=2}^{\infty} \frac{(\log m)^2}{m^3} \leq \frac{(\log 2)^2}{8} + \int_2^{\infty} \frac{(\log x)^2}{x^3} dx.$$

A direct calculation gives

$$\int_2^{\infty} \frac{(\log x)^2}{x^3} dx = \frac{(\log 2)^2 + \log 2 + \frac{1}{2}}{8}.$$

Hence

$$(\log \zeta)''(s) \leq \frac{2(\log 2)^2 + \log 2 + \frac{1}{2}}{8} < \frac{1}{3}.$$

□

5 Application I: Nantomah's zeta positivity problem

Nantomah [1] asked whether

$$K(n) := (n+2)\zeta(n+1)\zeta(n+3) - (n+1)\zeta(n+2)^2 - \zeta(n+1)\zeta(n+2)$$

is positive for all $n \in \mathbb{N}$. The zeta-law moment calculus gives an affirmative answer.

Theorem 8 (Affirmative solution of Nantomah's problem). *For every $n \in \mathbb{N}$,*

$$(n+2)\zeta(n+1)\zeta(n+3) - (n+1)\zeta(n+2)^2 - \zeta(n+1)\zeta(n+2) > 0. \quad (7)$$

Proof. Put

$$s = n+1 \geq 2, \quad Z_s = \zeta(s).$$

The expression becomes

$$\mathcal{K}_s = (s+1)Z_s Z_{s+2} - sZ_{s+1}^2 - Z_s Z_{s+1}.$$

Let $X(m) = 1/m$ under the zeta law ρ_s . By Proposition 1,

$$\mathbb{E}_s[X] = \frac{Z_{s+1}}{Z_s}, \quad \mathbb{E}_s[X^2] = \frac{Z_{s+2}}{Z_s}.$$

Therefore

$$\frac{\mathcal{K}_s}{Z_s^2} = (s+1) \mathbb{E}_s[X^2] - s \mathbb{E}_s[X]^2 - \mathbb{E}_s[X].$$

Equivalently,

$$\frac{\mathcal{K}_s}{Z_s^2} = (s+1) \text{Var}_s(X) + \mathbb{E}_s[X]^2 - \mathbb{E}_s[X].$$

This is the moment form of the problem.

We now prove positivity. Define

$$a = \zeta(s) - 1, \quad b = \zeta(s+1) - 1, \quad c = \zeta(s+2) - 1.$$

Then

$$Z_s = 1 + a, \quad Z_{s+1} = 1 + b, \quad Z_{s+2} = 1 + c.$$

Expanding $\mathcal{K}_s$ gives

$$\mathcal{K}_s = L_s + Q_s,$$

where

$$L_s = sa + (s+1)c - (2s+1)b$$

and

$$Q_s = (s+1)ac - sb^2 - ab.$$

The linear part is positive term by term:

$$\begin{aligned} L_s &= \sum_{m=2}^{\infty} m^{-s} \left(s - \frac{2s+1}{m} + \frac{s+1}{m^2} \right) \\ &= \sum_{m=2}^{\infty} m^{-s} \frac{(m-1)(s(m-1)-1)}{m^2}. \end{aligned}$$

For $s \geq 2$ and $m \geq 2$, the summand is positive. In particular, the $m=2$ term gives

$$L_s \geq 2^{-s} \frac{s-1}{4}.$$

For the quadratic correction, observe that

$$b = \sum_{m=2}^{\infty} m^{-s-1} \leq \frac{1}{2} \sum_{m=2}^{\infty} m^{-s} = \frac{a}{2}.$$

Since $c \geq 0$,

$$Q_s = (s+1)ac - sb^2 - ab \geq -sb^2 - ab \geq -\frac{s+2}{4}a^2.$$

For $s \geq 4$,

$$a = \sum_{m=2}^{\infty} m^{-s} \leq 2^{-s} + \int_2^{\infty} x^{-s} dx = 2^{-s} \left(1 + \frac{2}{s-1} \right).$$

Hence, for $s \geq 4$,

$$\mathcal{K}_s \geq \frac{1}{4} \left[2^{-s}(s-1) - (s+2)2^{-2s} \left(1 + \frac{2}{s-1} \right)^2 \right].$$

It is enough to prove

$$2^s(s-1) > (s+2) \left(1 + \frac{2}{s-1}\right)^2.$$

Equivalently,

$$F(s) := \frac{2^s(s-1)^3}{(s+2)(s+1)^2} > 1.$$

At $s = 4$,

$$F(4) = \frac{16 \cdot 27}{6 \cdot 25} = \frac{72}{25} > 1.$$

Moreover,

$$\frac{d}{ds} \log F(s) = \log 2 + \frac{3}{s-1} - \frac{1}{s+2} - \frac{2}{s+1} > 0 \quad (s \geq 4),$$

because

$$\frac{3}{s-1} - \frac{2}{s+1} - \frac{1}{s+2} > \frac{3}{s+1} - \frac{2}{s+1} - \frac{1}{s+2} = \frac{1}{s+1} - \frac{1}{s+2} > 0.$$

Thus $\mathcal{K}_s > 0$ for all $s \geq 4$.

It remains to check $s = 2$ and $s = 3$.

For $s = 2$,

$$\mathcal{K}_2 = 3\zeta(2)\zeta(4) - 2\zeta(3)^2 - \zeta(2)\zeta(3).$$

Using

$$\zeta(3) < 1 + \frac{1}{8} + \int_2^\infty x^{-3} dx = \frac{5}{4},$$

together with $\zeta(2) = \pi^2/6$ and $\zeta(4) = \pi^4/90$, we obtain

$$\mathcal{K}_2 > \frac{\pi^6}{180} - \frac{5\pi^2}{24} - \frac{25}{8}.$$

The function

$$f(x) = \frac{x^6}{180} - \frac{5x^2}{24} - \frac{25}{8}$$

is increasing for $x \geq 3$. Since $\pi > 313/100$,

$$\mathcal{K}_2 > f\left(\frac{313}{100}\right) = \frac{10415360504209}{18000000000000} > 0.$$

For $s = 3$,

$$\mathcal{K}_3 = 4\zeta(3)\zeta(5) - 3\zeta(4)^2 - \zeta(3)\zeta(4).$$

We use the elementary bounds

$$\zeta(3) > 1 + \frac{1}{8} + \frac{1}{27} = \frac{251}{216},$$

$$\zeta(5) > 1 + \frac{1}{32} + \frac{1}{243} = \frac{8051}{7776},$$

and

$$\zeta(4) = \frac{\pi^4}{90} < \frac{13}{12}.$$

Since

$$4 \cdot \frac{8051}{7776} - \frac{13}{12} > 0,$$

we get

$$\begin{aligned}\mathcal{K}_3 &> \frac{251}{216} \left(4 \cdot \frac{8051}{7776} - \frac{13}{12} \right) - 3 \left(\frac{13}{12} \right)^2 \\ &= \frac{13783}{419904} > 0.\end{aligned}$$

Therefore $\mathcal{K}_s > 0$ for every integer $s \geq 2$, and hence $K(n) > 0$ for every $n \in \mathbb{N}$. $\square$

6 Application II: Alzer–Kwong convexity and concavity problem

Alzer and Kwong [3, Conjecture 1.12] conjectured that, for every $n \in \mathbb{N}$, the function $1/\zeta$ is strictly convex on

$$(-4n, -4n + 2)$$

and strictly concave on

$$(-4n - 2, -4n).$$

The free-energy curvature bound above gives a direct proof.

Theorem 9 (Alzer–Kwong Conjecture 1.12). *Let*

$$F(x) = \frac{1}{\zeta(x)}.$$

For every integer $n \geq 1$,

$$F''(x) > 0 \quad \text{on} \quad (-4n, -4n + 2). \quad (8)$$

and

$$F''(x) < 0 \quad \text{on} \quad (-4n - 2, -4n). \quad (9)$$

Thus $1/\zeta$ is strictly convex on $(-4n, -4n + 2)$ and strictly concave on $(-4n - 2, -4n)$.

Proof. Put

$$x = -u, \quad u > 2.$$

The functional equation of the Riemann zeta function gives

$$\zeta(-u) = -2^{-u} \pi^{-u-1} \sin\left(\frac{\pi u}{2}\right) \Gamma(u+1) \zeta(u+1).$$

Therefore

$$\frac{1}{\zeta(-u)} = -\frac{2^u \pi^{u+1}}{\Gamma(u+1) \zeta(u+1) \sin(\pi u/2)}.$$

Define the positive function

$$G(u) := \frac{2^u \pi^{u+1}}{\Gamma(u+1) \zeta(u+1) |\sin(\pi u/2)|}.$$

On $u \in (4n - 2, 4n)$,

$$\sin\left(\frac{\pi u}{2}\right) < 0,$$

so

$$\frac{1}{\zeta(-u)} = G(u).$$

On $u \in (4n, 4n + 2)$,

$$\sin\left(\frac{\pi u}{2}\right) > 0,$$

so

$$\frac{1}{\zeta(-u)} = -G(u).$$

Since $x = -u$, second derivatives are unchanged:

$$\frac{d^2}{dx^2}F(x) = \frac{d^2}{du^2}F(-u).$$

It is therefore enough to prove

$$G'''(u) > 0 \quad (u > 2).$$

Let

$$\ell(u) = \log G(u).$$

Then

$$\ell(u) = u \log 2 + (u + 1) \log \pi - \log \Gamma(u + 1) - \log \zeta(u + 1) - \log \left| \sin\left(\frac{\pi u}{2}\right) \right|.$$

Differentiating twice,

$$\ell''(u) = -\psi'(u + 1) - (\log \zeta)''(u + 1) + \frac{\pi^2}{4} \csc^2\left(\frac{\pi u}{2}\right),$$

where ψ' is the trigamma function.

Put

$$s = u + 1.$$

Since $u > 2$, we have $s > 3$. The trigamma function satisfies

$$\psi'(s) = \sum_{k=0}^{\infty} \frac{1}{(s+k)^2} < \int_{s-1}^{\infty} \frac{dt}{t^2} = \frac{1}{s-1} \leq \frac{1}{2}.$$

By Lemma 7,

$$(\log \zeta)''(s) < \frac{1}{3}.$$

Also,

$$\csc^2\left(\frac{\pi u}{2}\right) \geq 1.$$

Therefore

$$\ell''(u) > \frac{\pi^2}{4} - \frac{1}{2} - \frac{1}{3} = \frac{\pi^2}{4} - \frac{5}{6} > 0.$$

Since $G > 0$,

$$G''(u) = G(u) (\ell''(u) + \ell'(u)^2) > 0.$$

Now return to $x = -u$.

If $x \in (-4n, -4n + 2)$, then $u \in (4n - 2, 4n)$, and

$$F(x) = F(-u) = G(u).$$

Thus

$$F''(x) = G''(u) > 0.$$

If $x \in (-4n - 2, -4n)$, then $u \in (4n, 4n + 2)$, and

$$F(x) = F(-u) = -G(u).$$

Thus

$$F''(x) = -G''(u) < 0.$$

This proves the claimed convexity and concavity pattern. $\square$

7 Application III: A generalized Hölder problem of Sroysang

Sroysang [2, Section 3] uses the notation

$$\xi(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{t^{s-1}}{e^t - 1} dt, \quad s > 1.$$

Thus, in standard notation,

$$\xi(s) = \zeta(s), \quad s > 1.$$

This is not Riemann's completed xi-function. In this section we write ζ in the main statement, and then restate the result in Sroysang's notation.

Sroysang proved the Hölder-type inequality

$$\zeta\left(\sum_{i=1}^m \frac{x_i}{p_i}\right) \leq \frac{\prod_{i=1}^m (\Gamma(x_i)\zeta(x_i))^{1/p_i}}{\Gamma(\sum_{i=1}^m x_i/p_i)},$$

under the assumptions

$$x_i > 1, \quad p_i > 1, \quad \sum_{i=1}^m \frac{1}{p_i} = 1.$$

He then asked how to generalize this inequality using the generalized Hölder condition

$$\sum_{i=1}^m \frac{1}{p_i} = \frac{1}{r}, \quad r \geq 1.$$

The following theorem gives the corresponding sharp extension.

Theorem 10 (Solution of Sroysang's generalized Hölder problem). *Let $m \geq 2$, let $r \geq 1$, and let*

$$x_1, \dots, x_m > 1, \quad p_1, \dots, p_m > 1,$$

satisfy

$$\sum_{i=1}^m \frac{1}{p_i} = \frac{1}{r}.$$

Define

$$T := r \sum_{i=1}^m \frac{x_i}{p_i}.$$

Then $T > 1$ and

$$\zeta(T) \leq \frac{\prod_{i=1}^m (\Gamma(x_i)\zeta(x_i))^{r/p_i}}{\Gamma(T)}. \quad (10)$$

Equivalently, in Sroysang's notation,

$$\xi\left(r \sum_{i=1}^m \frac{x_i}{p_i}\right) \leq \frac{\prod_{i=1}^m (\Gamma(x_i)\xi(x_i))^{r/p_i}}{\Gamma(r \sum_{i=1}^m x_i/p_i)}. \quad (11)$$

For $r = 1$, this reduces to Sroysang's inequality.

Proof. Let

$$M(s) := \Gamma(s)\zeta(s) = \int_0^\infty \frac{t^{s-1}}{e^t - 1} dt, \quad s > 1.$$

Set

$$\lambda_i := \frac{r}{p_i}.$$

Then

$$\lambda_i > 0, \quad \sum_{i=1}^m \lambda_i = 1,$$

and

$$T = \sum_{i=1}^m \lambda_i x_i.$$

Since each $x_i > 1$, we have $T > 1$.

We prove the inequality directly from generalized Hölder. Define

$$q_i := \frac{p_i}{r}.$$

Then

$$\sum_{i=1}^m \frac{1}{q_i} = 1.$$

For $m \geq 2$, the assumptions imply $q_i > 1$ for all i . Indeed, if $q_j \leq 1$ for some j , then $1/p_j \geq 1/r$, leaving no room for the other positive terms in $\sum_i 1/p_i = 1/r$.

Now define

$$f_i(t) := \left(\frac{t^{x_i-1}}{e^t - 1}\right)^{1/p_i}, \quad t > 0.$$

Then

$$\prod_{i=1}^m f_i(t) = t^{\sum_i (x_i-1)/p_i} (e^t - 1)^{-\sum_i 1/p_i} = t^{\sum_i (x_i-1)/p_i} (e^t - 1)^{-1/r}.$$

Therefore

$$\left(\prod_{i=1}^m f_i(t)\right)^r = \frac{t^{r \sum_i (x_i-1)/p_i}}{e^t - 1}.$$

Using

$$r \sum_{i=1}^m \frac{x_i - 1}{p_i} = r \sum_{i=1}^m \frac{x_i}{p_i} - r \sum_{i=1}^m \frac{1}{p_i} = T - 1,$$

we get

$$\left(\prod_{i=1}^m f_i(t)\right)^r = \frac{t^{T-1}}{e^t - 1}.$$

Hence

$$\left\| \prod_{i=1}^m f_i \right\|_{L^r(0,\infty)}^r = \int_0^\infty \frac{t^{T-1}}{e^t - 1} dt = \Gamma(T)\zeta(T).$$

By generalized Hölder,

$$\left\| \prod_{i=1}^m f_i \right\|_{L^r} \leq \prod_{i=1}^m \|f_i\|_{L^{p_i}}.$$

But

$$\|f_i\|_{L^{p_i}}^{p_i} = \int_0^\infty \frac{t^{x_i-1}}{e^t - 1} dt = \Gamma(x_i)\zeta(x_i).$$

Therefore

$$(\Gamma(T)\zeta(T))^{1/r} \leq \prod_{i=1}^m (\Gamma(x_i)\zeta(x_i))^{1/p_i}.$$

Raising both sides to the power r gives

$$\Gamma(T)\zeta(T) \leq \prod_{i=1}^m (\Gamma(x_i)\zeta(x_i))^{r/p_i}.$$

Dividing by $\Gamma(T)$ proves the result. $\square$

Remark 11 (Free-energy interpretation). *The same result can be phrased as log-convexity of the Mellin–Planck partition function*

$$M(s) = \Gamma(s)\zeta(s) = \int_0^\infty \frac{t^{s-1}}{e^t - 1} dt.$$

For $s > 1$, define the probability law

$$\nu_s(dt) := \frac{t^{s-1}}{(e^t - 1)M(s)} dt.$$

Then

$$\frac{d^2}{ds^2} \log M(s) = \text{Var}_{\nu_s}(\log t) \geq 0.$$

Thus M is log-convex on $(1, \infty)$. With

$$\lambda_i = \frac{r}{p_i}, \quad \sum_{i=1}^m \lambda_i = 1, \quad T = \sum_{i=1}^m \lambda_i x_i,$$

Jensen's inequality gives

$$M(T) \leq \prod_{i=1}^m M(x_i)^{\lambda_i} = \prod_{i=1}^m (\Gamma(x_i)\zeta(x_i))^{r/p_i}.$$

Since $M(T) = \Gamma(T)\zeta(T)$, this is exactly Theorem 10. This places Sroysang's generalized Hölder problem inside the same zeta-law free-energy framework used throughout the paper.

8 Summary of the framework

The results above use four levels of the same zeta-law viewpoint.

First, the moment layer:

$$\frac{\zeta(s+r)}{\zeta(s)} = \mathbb{E}_s[N^{-r}].$$

This converts zeta-ratio inequalities into moment inequalities under the Riemann law. Theorem 8 uses this layer.

Second, the free-energy curvature layer:

$$(\log \zeta)''(s) = \text{Var}_s(\log N).$$

This converts positive-axis control of ζ into thermodynamic curvature estimates. Theorem 9 uses this layer, transported to the negative axis by the functional equation.

Third, the Mellin–Planck free-energy layer:

$$M(s) = \Gamma(s)\zeta(s), \quad (\log M)''(s) = \text{Var}_{\nu_s}(\log t).$$

This converts generalized Hölder inequalities into log-convexity of a continuous zeta partition function. Theorem 10 uses this layer.

Fourth, the modular entropy layer:

$$\Sigma(\beta) = \sup_{q \geq 2} B_q(\beta).$$

This says that additive successor entropy of the microscopic zeta law is recovered from finite modular observations. For prime moduli, these modular observations are explicit nonlinear functionals of Dirichlet L -values.

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$$(n+2)\zeta(n+1)\zeta(n+3) - (n+1)\zeta(n+2)^2 - \zeta(n+1)\zeta(n+2)$$

is positive for all $n \in \mathbb{N}$.

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- [3] Horst Alzer and Man Kam Kwong, *On the concavity and convexity of $1/\zeta$* , *International Journal of Number Theory*, Vol. 21, No. 8 (2025), 1825–1835. DOI: <https://doi.org/10.1142/S1793042125500897>. Conjecture 1.12 asks for the convexity and concavity pattern of $1/\zeta$ on the intervals $(-4n, -4n+2)$ and $(-4n-2, -4n)$.
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